



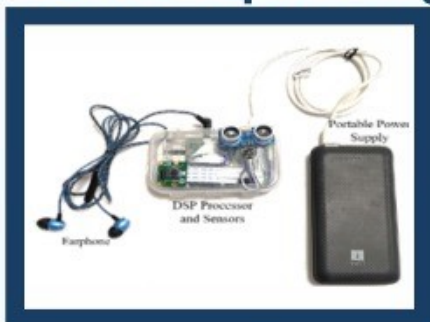
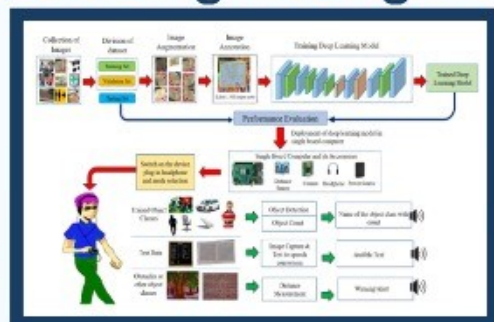
CENTRE FOR ADVANCED STUDIES

LUCKNOW, UTTAR PRADESH, INDIA

INSPIRING STUDENT INNOVATION

Assistive Device to Impart Perceptual Ability to Visually-Impaired using Intelligent Scene Captioning

A team of Centre for Advanced Studies developed an Artificial Intelligence based fully automatic assistive technology to recognize different objects, and auditory inputs are provided to the user in real time, which gives better understanding to the visually impaired person about their surroundings. A deep-learning model is



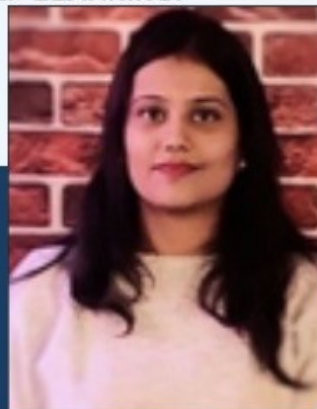
trained with multiple images of objects that are highly relevant to the visually impaired person. Training images are augmented and manually annotated to bring more robustness to the trained model. In addition to computer vision-based techniques for object recognition, a distance-measuring sensor is integrated to make the device more comprehensive by recognizing obstacles while navigating from one place to another. The auditory information that is conveyed to the user after scene segmentation and obstacle identification is optimized to obtain more information in less time for faster processing of video frames. The whole framework is standalone and designed for low-cost processors, so that the visually impaired can use it properly without internet connectivity. The detected output is also optimized to ensure faster frame processing and a greater extraction of information—i.e., count of objects—in a shorter time period. The proposed system can easily differentiate between obstacles and known objects.

THIS INNOVATION OF CENTRE FOR ADVANCED STUDIES CAN MAKE A SIGNIFICANT CONTRIBUTION TO ASSIST VISUALLY IMPAIRED PEOPLE COMPARED TO PREVIOUSLY DEVELOPED METHODS, WHICH WERE ONLY FOCUSED ON OBSTACLE DETECTION AND LOCATION TRACKING WITH THE HELP OF BASIC SENSORS WITHOUT USE OF DEEP LEARNING.

RESEARCH TEAM



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PATENTS FROM THE PROJECT:

Multi Mode Real-Time Object Detector and Obstacle Aware Assistive System for Visually Impaired Person (VIP)", Application No.- 201911047766,
Dated - 22.11.2019, Status – Applied and under examination.

RESEARCH PUBLICATION FROM THIS WORK

1. Rakesh Chandra Joshi, Saumya Yadav, Malay Kishore Dutta, Carlos M. Travieso-Gonzalez, "An Efficient Multi-Object Detection and Smart Navigation using Artificial Intelligence for Visually Impaired People" Entropy, Volume 22 Issue 9, 2020, DOI :10.3390/e22090941, SCI indexed Impact Factor – 2.419.
2. Rakesh Chandra Joshi, Saumya Yadav & Malay Kishore Dutta, "YOLO-v3 Based Currency Detection and Recognition System for Visually Impaired Persons" International Conference on Contemporary Computing, 2020, IEEE Xplore Digital Library, DOI:10.1109/IC3A48958.2020.233314.
3. Saumya Yadav, Rakesh Chandra Joshi, Malay Kishore Dutta, Martin Kiac, Pavel Sikora, "Fusion of Object Recognition and Obstacle Detection approach for Assisting Visually Challenged Person" 43rd International Conference on Telecommunications and Signal Processing (TSP), 2020, Italy, IEEE Xplore digital library, DOI: 10.1109/TSP49548.2020.9163434